R1-1 分数 2 作者 陈越 单位 浙江大学

All decidable problems are NP problems.

- T
- F

R1-2 分数 2 作者 叶德仕 单位 浙江大学

In local search, if the optimization function has a constant value in a neighborhood, there will be a problem.

- T
- F

R1-3 分数 2 作者 陈越 单位 浙江大学

Finding the minimum key from a splay tree will result in a tree with its root having no left subtree.

- T
- F

R1-4 分数 2 作者 陈越 单位 浙江大学

If a problem can be solved by dynamic programming, it must be solved in polynomial time.

- T
- F

R1-5 分数 2 作者 陈越 单位 浙江大学

For the recurrence equation $T(N) = aT(N/b) + f(N)$, if $af(N/b) = f(N)$, then $T(N) = \Theta(Nlog_bN)$.

- T
- F

R1-6 分数 2 作者 陈越 单位 浙江大学

An approximation scheme that runs in $O(n^2/\epsilon)$ for any fixed $\epsilon > 0$ is a fully polynomial-time approximation scheme.

- T
- F

R1-7 分数 2 作者 叶德仕 单位 浙江大学

Consider the online hiring problem, in which we have total k candidates. First of all, we interview n candidates but reject them all. Then we hire the first candidate who is better than all of the previous candidates you have interviewed. It is true that the probability of the mth candidate is the best is $\frac{n}{k(m-1)}$, where $m > n$.

- T
- F

R1-8 分数 2 作者 陈越 单位 浙江大学

For any node in an AVL tree, the height of the right subtree must be greater than that of the left subtree.

- T
- F

R1-9 分数 2 作者 陈越 单位 浙江大学

For a binomial queue, delete-min takes a constant time on average.

- T
- F

R1-10 分数 2 作者 陈越 单位 浙江大学

With the same operations, the resulting skew heap is always more balanced than the leftist heap.

- T
- F

R1-11 分数 2 作者 陈越 单位 浙江大学

In amortized analysis, a good potential function should always assume its maximum at the start of the sequence.

- T
- F

R1-12 分数 2 作者 徐镜春 单位 浙江大学

Let S be the set of activities in Activity Selection Problem. The greedy rule of "collecting the activity that starts the latest" is correct for finding a maximum-size subset of mutually compatible activities of S.

- T
- F

R1-13 分数 2 作者 何钦铭 单位 浙江大学

For any red node X in a Red-Black tree, if it has two children, then the children's colors must be the same.

- T
- F

R1-14 分数 2 作者 陈越 单位 浙江大学

The time bound of the FIND operation in a B+ tree containing N numbers is $O(\log N)$, no matter what the degree of the tree is.

- T
- F

R1-15 分数 2 作者 杨欣豫 单位 浙江大学

While accessing a term by hashing in an inverted file index, range searches are expensive.

- T
- F

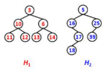
R2-1 分数 3 作者 叶德仕 单位 浙江大学

In the bin packing problem, suppose that the maximum size of the items is bounded from above by $\alpha < 1$. Now let's apply the Next Fit algorithm to pack a set of items L into bins with capacity 1. Let $NF(L)$ and $OPT(L)$ denote the numbers of bins used by algorithms Next Fit and the optimal solution. Then for all L, we have $NF(L) < \rho \cdot OPT(L) + 1$ for some ρ. Which one of the following statements is FALSE?

- A. $\rho = \frac{1}{1-\alpha}$, if $0 \le \alpha \le 1/2$.
- B. $\rho = 2$, if $\alpha > 1/2$.
- C. If L = (0.5, 0.3, 0.4, 0.8, 0.2, 0.3), then $NF(L) = 4$.
- D. $\rho = 2\alpha$ if $0 \le \alpha \le 1$

R2-2 分数 3 作者 陈越 单位 浙江大学

Merge the two given skew heaps. Which one of the following statements is FALSE?



- A. 25 is the left child of 14
- B. 39 is the right child of 25
- C. 3 is the root
- D. 17 is the right child of 16

R2-3 分数 3 作者 陈越 单位 浙江大学

Which of the following binomial trees can represent a binomial queue of size 78?

- A. $B_1\ B_2\ B_3\ B_6$
- B. $B_1\ B_8$
- C. $B_1\ B_2\ B_3\ B_5\ B_5$
- D. $B_0\ B_1\ B_4\ B_5\ B_6$

R2-4 分数 3 作者 Can Wang 单位 浙江大学

```
If X is a problem in class NP, then how many of the following statements is/are TRUE?
```

- There is no polynomial time algorithm for X.
- There is a polynomial time algorithm for X.
- If X can be solved deterministically in polynomial time, then P = NP.

- A. 0
- B. 3
- C. 2
- D. 1

R2-5 分数 3 作者 卜佳俊 单位 浙江大学

Insert { 9, 8, 7, 2, 3, 5, 6, 4 } one by one into an initially empty AVL tree. How many of the following statements is/are FALSE?

- the total number of rotations made is 5 (Note: double rotation counts 2 and single rotation counts 1)
- the expectation (round to 0.01) of access time is 2.77
- there are 2 nodes with a balance factor of -1

- A. 3
- B. 1
- C. 2
- D. 0

R2-6 分数 3 作者 徐镜春 单位 浙江大学

Given 4 characters (u, v, w, x) with some frequencies (f_u, f_v, f_w, f_x) in a text. If the corresponding Huffman codes are u: 00, v: 010, w: 011 and x: 1. Which of the following sets of frequencies is a possible one for (f_u, f_v, f_w, f_x) ?

- A. 55, 22, 18, 46
- B. 41, 12, 20, 32
- C. 30, 21, 12, 33
- D. 15, 23, 16, 45

R2-7 分数 3 作者 何钦铭 单位 浙江大学

A replacement selection is applied to generate the max run with a priority queue of 5 records. When the sequence of numbers is { 6, 79, 14, 10, 90, 28, 35, X, ... } and the length of the first run is 7, what is the sufficient condition of X?

- A. less than 90
- B. less than 35
- C. greater than 90
- D. less than 14

R2-8 分数 3 作者 沈鑫 单位 浙江大学

Given a linked list containg N nodes. Our task is to remove all the nodes. At each step, we randomly choose one node in the current list, then delete the selected node together with all the nodes after it. Here we assume that each time we choose one node uniformly among all the remaining nodes. What is the expected number of steps to remove all the nodes?

- A. $N/2$
- B. $\sqrt{N}$
- C. N/e
- D. $\Theta(logN)$

R2-9 分数 3 作者 陈越 单位 浙江大学

Insert { 5, 1, 7, 8, 21, 2, 12, 19, 13, 0 } into an initially empty 2-3 tree (with splitting). Which one of the following statements is FALSE?

- A. there are 5 leaf nodes
- B. 13 and 19 are in the same node
- C. the parent of the node containing 8 has 3 children
- D. the first key stored in the root is 12

R2-10 分数 3 作者 张国川 单位 浙江大学

Consider the following buffer management problem. Initially the buffer size (the number of blocks) is one. Each block can accommodate exactly one item. As soon as a new item arrives, check if there is an available block. If yes, put the item into the block, induced a cost of one. Otherwise, the buffer size is doubled, and then the item is able to put into. Moreover, the old items have to be moved into the new buffer so it costs $k+1$ to make this insertion, where k is the number of old items. Clearly, if there are N items, the worst-case cost for one insertion can be $\Omega(N)$. To show that the average cost is $O(1)$, let us turn to the amortized analysis. To simplify the problem, assume that the buffer is full after all the N items are placed. Which of the following potential functions works?

- A. The number of available blocks currently in the buffer
- B. The number of items currently in the buffer
- C. The opposite number of items currently in the buffer
- D. The opposite number of available blocks in the buffer

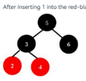
R2-11 分数 3 作者 徐镜春 单位 浙江大学

When solving a problem with input size N by divide and conquer, if at each step, the problem is divided into 4 sub-problems of size $\sqrt{N}$, and they are conquered in $O(logN)$. Which one of the following is the closest to the overall time complexity $T(N)$? Suppose that $T(2) = O(1)$ and all related root values are integers.

- A. $O(log^2 N)$
- B. $O(loglogN)$
- C. $O(logN)$
- D. $O(N)$

R2-12 分数 3 作者 何钦铭 单位 浙江大学

After inserting 1 into the red-black tree given in the figure, which node(s) will have its/their color(s) changed?



- A. both 2 and 4
- B. both 3 and 6
- C. 2, 3, and 4
- D. only 2

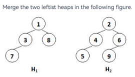
R2-13 分数 3 作者 沈鑫 单位 浙江大学

Which one of the following statements about the Maximum Finding problem is true?

- A. No parallel algorithm can solve the problem in $O(1)$ time.
- B. When partitioning the problem into sub-problems and solving them in parallel, compared with $\sqrt{N}$, choosing $loglogN$ as the size of each sub-problem can reduce the work load and the worst-case time complexity.
- C. There exists a serial algorithm with time complexity being $O(logN)$.
- D. Parallel random sampling algorithm can run in $O(1)$ time and $O(N)$ work with very high probability.

R2-14 分数 3 作者 陈越 单位 浙江大学

Merge the two leftist heaps in the following figure. Which one of the following statements is FALSE?



- A. 6 is the left child of 2
- B. 1 is the root with 3 being its right child
- C. the null path length of 6 is the same as that of 2
- D. Along the left most path from top down, we have 1, 2, 4, and 5

R2-15 分数 3

Spanning Tree Problem: Given an undirected graph $G=(V,E)$, where $|V|=n$ and $|E|=m$. Let F be the set of all spanning trees of G. Define $d(u)$ to be the degree of a vertex $u \in V$. Define $w(e)$ to be the weight of an edge $e \in E$.
We have the following three variants of spanning tree problems:

- (1) Max Leaf Spanning Tree: find a spanning tree $T \in F$ with a maximum number of leaves.
- (2) Minimum Spanning Tree: find a spanning tree $T \in F$ with a minimum total weight of all the edges in T.
- (3) Minimum Degree Spanning Tree: find a spanning tree $T \in F$ such that its maximum degree of all the vertices is the smallest.

For a pair of edges (e, e') where $e \in T$ and $e' \in (G-T)$ such that e belongs to the unique cycle of $T \cup e'$, we define **edge-swap**(e, e') to be $(T-e) \cup e'$.

Here is a local search algorithm:

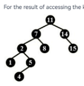
```
T = any spanning tree in F;
while (there is an edge-swap(e, e') which reduces Cost(T)) {
    T = T - e + e';
}
return T;
```

Here `Cost(T)` is the number of leaves in `T` in Max Leaf Spanning Tree; or is the total weight of `T` in Minimum Spanning Tree; or else is the minimum degree of `T` in Minimum Degree Spanning Tree.
Which of the following statements is TRUE?

- A. For neither of the problems that this local search always return an optimal solution
- B. The local search always return an optimal solution for Minimum Degree Spanning Tree
- C. The local search always return an optimal solution for Minimum Spanning Tree
- D. The local search always return an optimal solution for Max Leaf Spanning Tree

R2-16 分数 3

For the result of accessing the keys 4 and 8 in order in the splay tree given in the figure, which one of the following statements is FALSE?



- A. 4 and 11 are siblings
- B. 7 and 14 are siblings
- C. 8 is the root
- D. 4 is the parent of 7

R2-17 分数 3

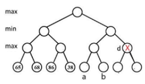
There are 8000 documents in the database. The statistic data for one query are shown in the following table. The precision is: __

	Relevant	Irrelevant
Retrieved	1000	1000
Not Retrieved	2000	4000

- A. 12.5%
- B. 50%
- C. 20%
- D. 33%

R2-18 分数 3

Given the following game tree, if node **d** is pruned with α-β pruning algorithm, which of the following statements about the value of node **a** or node **b** is correct?



- A. one of them greater than 68
- B. both are less than 68
- C. both are greater than 68
- D. one of them less than 68

R2-19 分数 3

Given a recurrence equation $f_{i,j,k} = f_{i,j+1,k} + \min_{0 \le l < k}\{f_{i-1,j,l} + w_{j,l}\}$. To solve this equation in an iterative way, we **cannot** fill up a table as follows:

- A. `for i in 0 to n: for j in n to 0: for k in 0 to n`
- B. `for i in 0 to n: for j in n to 0: for k in n to 0`
- C. `for i in 0 to n: for j in 0 to n: for k in 0 to n`
- D. `for k in 0 to n: for i in 0 to n: for j in n to 0`

R5-1 分数 6 BinQueue_DeleteRoot

The function `DeleteRoot` is to delete the root of a subtree with index `Ind` from a binomial queue `H`. The rest of the subtree is then stored as a new binomial queue and returned.

```
BinQueue DeleteRoot( BinQueue H, int Ind )
{
    BinTree OldRoot, SubTree;
    BinQueue NewBinQ;
    int i;

    OldRoot = H->TheTrees[Ind];
    SubTree = OldRoot->LeftChild;
    free(OldRoot);
    NewBinQ = Initialize();
    NewBinQ->CurrentSize = ________ 3分;
    for ( ________ 3分 ) {
        NewBinQ->TheTrees[i] = SubTree;
        SubTree = SubTree->NextSibling;
        NewBinQ->TheTrees[i]->NextSibling = NULL;
    }
    return NewBinQ;
}
```

R6-1 Decode 分数 7

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作者 陈越 单位 浙江大学

Suppose that a string of English letters is encoded into a string of numbers. To be more specific, `A`-`Z` are encoded into `0`-`25`. Since it is not a prefix code, the decoded result may not be unique. For example, `1213407` can be decoded as `BCBDEAH`, `MBDEAH`, `BCNEAH`, `BVDEAH` or `MNEAH`. Note that `07` is not `7`, hence cannot be decoded as `H`.

Your job is to tell in how many different ways we can decode a numeric string.

Format of function:

```
int Decode( char NumStr[] );
```

where `NumStr` is a string consisting of only the numbers `0`-`9`.

The function `Decode` is supposed to return the number of different ways we can decode `NumStr`.

Since the answer might be super large, you only need to output the answer modulo 1000000007.

Sample program of judge:

```
#include <stdio.h>
#include <string.h>

#define MAXN 100
#define BASE 1000000007

int Decode( char NumStr[] );

int main()
{
    char NumStr[MAXN];

    scanf("%s", NumStr);
    printf("%d", Decode(NumStr));

    return 0;
}

/* Your function will be put here */
```

Sample Input:

```
1213407
```

Sample Output:

```
5
```

代码长度限制	16 KB
时间限制	100 ms
内存限制	64 MB